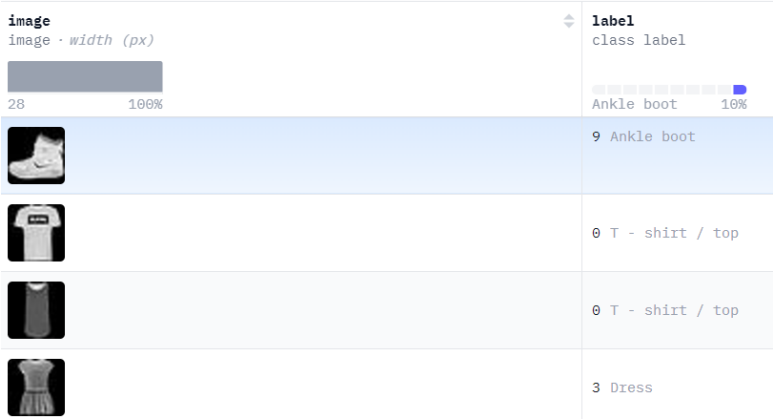
人工智能原理2025课程实验

课程实验-提交要求

- 命名格式：人工智能原理2025课程实验+学号+姓名
- 实验结果保存为“**experiment_X_test.csv**”，格式如下图，**第一列是id，按升序排列**
- 文件要求：zip格式的压缩包，包含必要的代码，实验结果，实验报告等内容，**不要提交数据集和模型**
- 截止时间：12月13日
- 发送邮箱： 2024110498@mail.hfut.edu.cn
- 实验报告：实验报告请参考评分准则，尽量提交完整的实验报告，实验报告命名方式要与压缩包保持一致，建议提交pdf，doc，docx格式
- 除实验结果格式不同外，实验一、实验二、实验三和实验四的提交要求一致。



课程实验一--住房价格预测

- **实验场景：**波士顿房地产市场竞争激烈，而你想成为该地区最好的房地产经纪人。帮助客户为自己的房产定下最佳售价。波士顿地区的房价受诸多因素影响。幸运的是，你找到了波士顿房价的数据集，里面聚合了波士顿郊区包含多个特征维度的房价数据。
- **实验任务：**你的任务是用可用的工具进行统计分析，并基于分析建立优化模型。这个模型将用来进行房价预测(MEDV)。

No	属性	数据类型	字段描述
1	CRIM	Float	城镇人均犯罪率
2	ZN	Float	占地面积超过2.5万平方英尺的住宅用地比例
3	INDUS	Float	城镇非零售业务地区的比例
4	CHAS	Integer	查尔斯河虚拟变量 (= 1 如果土地在河边；否则是0)
5	NOX	Float	一氧化氮浓度（每1000万份）
6	RM	Float	平均每居民房数
7	AGE	Float	在1940年之前建成的所有者占用单位的比例
8	DIS	Float	与五个波士顿就业中心的加权距离
9	RAD	Integer	辐射状公路的可达性指数
10	TAX	Float	每10,000美元的全额物业税率
11	PTRATIO	Float	城镇师生比例
12	B	Float	1000（Bk - 0.63）^ 2其中Bk是城镇黑人的比例
13	LSTAT	Float	人口中地位较低人群的百分数
14	MEDV	Float	（目标变量/类别属性）以1000美元计算的自有住房的中位数

课程实验一--住房价格预测

- 数据集：
  - 提供按照8:2划分Train, Test数据集
  - Train提供’MEDV’标签，**测试集不提供，需要进行预测**

Train

CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
0.05188	0	4.49	0	0.449	6.015	45.1	4.4272	3	247	18.5	395.99	12.86	22.5
0.0459	52.5	5.32	0	0.405	6.315	45.6	7.3172	6	293	16.6	396.9	7.6	22.3
13.5222	0	18.1	0	0.631	3.863	100	1.5106	24	666	20.2	131.42	13.33	23.1
0.77299	0	8.14	0	0.538	6.495	94.4	4.4547	4	307	21	387.94	12.8	18.4
0.05735	0	4.49	0	0.449	6.63	56.1	4.4377	3	247	18.5	392.3	6.53	26.6
0.04337	21	5.64	0	0.439	6.115	63	6.8147	4	243	16.8	393.97	9.43	20.5
0.11132	0	27.74	0	0.609	5.983	83.5	2.1099	4	711	20.1	396.9	13.35	20.1
0.03961	0	5.19	0	0.515	6.037	34.5	5.9853	5	224	20.2	396.9	8.01	21.1

模型预测

Test

idx	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
0	9.32909	0	18.1	0	0.713	6.185	98.7	2.2616	24	666	20.2	396.9	18.13	13.7
1	2.14918	0	19.58	0	0.871	5.709	98.5	1.6232	5	403	14.7	261.95	15.79	16.5
2	2.24236	0	19.58	0	0.605	5.854	91.8	2.422	5	403	14.7	395.11	11.64	19.3
3	0.31827	0	9.9	0	0.544	5.914	83.2	3.9986	4	304	18.4	390.7	18.33	14.7
4	0.01439	60	2.93	0	0.401	6.604	18.8	6.2196	1	265	15.6	376.7	4.38	15.6
5	0.41238	0	6.2	0	0.504	7.163	79.9	3.2157	8	307	17.4	372.08	6.36	20.1

课程实验一--住房价格预测

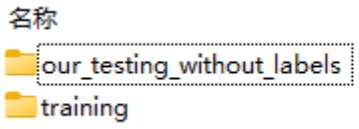
- 评价指标

$$RMSE = \frac{1}{n}\sum_{i=0}^{n}(Y_{预测i} - Y_{真实i})^2$$

- 实验要求：
  - 训练时只能利用train文件中的数据，优化好后，再进行测试集预测
    - 任务输入：波士顿房子的各类属性数据
    - 任务输出：设计模型，评估MEDV
  - 使用实验test.csv来测试训练好的模型，需要填充模型预测的自有住房的房价中位数到MEDV列。为方便测试，提交的test.csv文件需要按照idx列升序排列。
  - 需要对实验结果进行分析
  - 提交的内容：
    - 代码 + house_test.csv + 实验报告
    - 无需提交训练好的模型（包括预训练的模型）

课程实验二--fashion衣物识别

- 实验目标
  - 了解卷积神经网络（CNN）的基础原理
  - 掌握如何使用Python和机器学习库（如TensorFlow或PyTorch）进行简单的图像分类
  - 理解深度学习模型的训练、验证和测试过程
- 实验内容
  - 使用Fashion数据集，它包含了很多服饰图片
  - 利用卷积神经网络（CNN）构建一个简单的模型，让它可以“学会”识别这些服饰
  - 在训练数据集集上训练模型
  - 用测试数据集来评估模型



测试数据集，不包含标签

训练数据集，包含图像和标签

课程实验三--MBTI

- 背景：
  - MBTI（Myers-Briggs Type Indicator）是一种广泛使用的性格分类工具，基于瑞士心理学家卡尔·荣格（Carl Jung）的心理学
    - MBTI通过一系列问题来评估个体在四个维度上的偏好：
      - 外向(E)与内向(I)、感觉(S)与直觉(N)、思考(T)与情感(F)、判断(J)与知觉(P)
      - 每个维度上的偏好组合起来，形成16种不同的性格类型，如INTJ、ENFP等
    - MBTI的目的在于帮助人们更好地理解自己和他人，提高个人和团队的效率。它在职业规划、团队建设、领导力发展等领域有着广泛的应用。
- 实验内容
  - 现在有一系列用户在论坛的推文，请根据推文内容推断出每个用户的MBTI类型

https://www.16personalities.com/free-personality-test

课程实验三--MBTI

- 评价指标
  - Accuracy：模型预测正确的结果所占的比例
- 实验要求：
  - 训练时只能利用train文件中的数据，用验证集选择最优的epoch，再进行测试集预测
  - 使用实验test.csv来测试训练好的模型，需要填充模型预测的MBTI类型到type列。为方便测试，提交的test.csv文件需要按照idx列升序排列。
  - 可以采用任意的深度学习方法实现(Word2Vec, FastText, Bert ..)
  - 需要对实验结果进行分析
  - 提交的内容：
    - 代码 + mbti_test.csv + 实验报告
    - 无需提交训练好的模型（包括预训练的模型）

Word2Vec: Efficient Estimation of Word Representations in Vector Space
FastText: Bag of Tricks for Efficient Text Classification
Bert: Pre-training of deep bidirectional transformers for language understanding

课程实验三--MBTI

- 数据集：
  - Posts包括近期用户推文内容，每个推文按照 `|||` 隔开
  - 提供按照7:1:2进行划分的训练集，验证集和测试集
  - Train和valid提供type标签，**测试集不提供，需要进行预测**

Train/valid

type	posts
ENFP	Its a bizarre condition of mine, gets me into trouble|||Does this happen
ISTP	'wonder are they huggable.. :unsure: I'll try to figure it out :tongue: and
ENTJ	'Hi ! There is some material in your answer to speculate about your cog
ENFP	'This is great and makes a lot of sense, I feel like each of these really to
INFJ	'so he is not Mr Right? I mean, I thought INFJ's know when they found

Test

idx	type	posts
0	INFP	'http://www.cartoonstock.com/newscartoons/cartoonists/mba/lowr
1	INFP	'https://www.youtube.com/watch?v=BuV4qXCqKrQ -------|||^^
2	ESTP	'Nope, I was never bullied, which I think is kind of weird, to be hon
3	ISFP	'I don't mind authority itself if it's doing a good job... but if I feel lik
4	模型预测	'https://images.moviepilot.com/images/c_fill,h_282,w_500/t_mp_qu

课程实验四--基于 BERT 的文本情感分类微调实验

- **实验背景：** 在基于文本的评论中，例如淘宝评论区等，我们需要根据文本预测该段文本所附带的情绪：积极 or 消极，从而判定是否为好评，因此我们需要对文本进行情感分析。
- **实验目标：** 实践使用大规模预训练语言模型（BERT）通过指令微调快速适应下游任务，理解少样本学习和高效率模型开发的原理。
- **实验内容：** 使用微调后的Bert模型，判断一段文本的情绪。

课程实验四--基于 BERT 的文本情感分类微调实验

- 数据集：
  - 提供按照8:2划分Train, Test数据集
  - Train提供标签，**测试集不提供，需要进行预测**

Train

	A	B
1	text	label
2	I grew up (b. 1965) watching and loving the Thunderbirds. All my mates at school watched. We play	0
3	When I put this movie in my DVD player, and sat down with a coke and some chips, I had some exp	0
4	Why do people who do not know what a particular time in the past was like feel the need to try to	0
5	Even though I have great interest in Biblical movies, I was bored to death every minute of the movie	0
6	Im a die hard Dads Army fan and nothing will ever change that. I got all the tapes, DVD's and audic	1
7	A terrible movie as everyone has said. What made me laugh was the cameo appearance by Scott M	0
8	Finally watched this shocking movie last night, and what a disturbing mindf**ker it is, and unbelieval	1
9	I caught this film on AZN on cable. It sounded like it would be a good film, a Japanese "Green Card	0
10	It may be the remake of 1987 Autumn's Tale after eleven years, as the director Mabel Cheung claim	1
11	My Super Ex Girlfriend turned out to be a pleasant surprise for me, I was really expecting a horrible	1
12	I can't believe people are looking for a plot in this film. This is Laural and Hardy. Lighten up already.	1
13	If you haven't seen the gong show TV series then you won't like this movie much at all, not that kno	0
14	I have always been a huge fan of "Homicide: Life On The Street" so when I heard there was a reunic	1
15	Greg Davis and Bryan Daly take some crazed statements by a terrorists, add some commentary by	0
16	A half-hearted attempt to bring Elvis Presley into the modern day, but despite a sexy little shower s	0
17	If you want a fun romp with loads of subtle humor, then you will enjoy this flick. I don't	1
18	I really wanted to be able to give this film a 10. I've long thought it was my favorite of the four mod	1
19	The main problem with "Power" is that it features way too may pointless characters and subplots th	0
20	The folks at Disney have a lot to explain. First and foremost, why anyone thought this lesser-sitcom	0
21	A friend told me of John Fante last summer after we got into a conversation about Charles Bukowsl	0
22	Ever since I heard of the Ralph Bakshi version of "The Lord of the Rings" I wondered: What the hell is	1
23	I sat through this film and i have to say it only just managed to keep my attention. The film would h	0
24	I don't care if some people voted this movie to be bad. If you want the Truth this is a Very Good M	1
25	I never really understood the controversy and hype this movie caused. Especially in French and the	0
26	Sweet, entertaining tale of a young 17 1/2 year old boy, controlled by by an overbearing religious r	1
27	First, I should mention that I really enjoyed ISHII Katsuhito's previous film "Samehada Otoko to Mon	0
28	The Argentinian music poet, Atahualpa Yupanqui, once said that some folk music repeats similarly a	1
29	I saw this film by chance on the small box. It has a fantastic and chilling scene about poisonous gas.	1
30	The small California town of Diablo is plagued with mysterious deaths after sheriff Robert Lopez une	0
31	It's good to see that Vintage Film Buff have correctly categorized their excellent DVD release as a "r	1
32	I saw this film via one of the actors' agents, and it surely conforms with a great deal that comes out	1
33	During the Civil War, there were many cases of divided loyalties; obviously, many occurred "In the B	0
34	How could they take such a beautifully animated gem like Don Bluth's All Dogs go to Heaven and b	0
35	First off let me say that this is probably in my opinion one of the 10 most underrated movies since t	1
36	In a way this is the disaster Fellini has been working towards all his life. The line between absurd mas	0
37	Three writers made a valiant attempt to adapt Jane Stanton Hitchcock's novel for the tube, yet this	0

Test

	A
1	text
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37	Three writers made a valiant attempt to adapt Jane Stanton Hitchcock's novel for the tube, yet this

课程实验四--基于 BERT 的文本情感分类微调实验

- 评价指标
  - 准确率Accuracy、精确率Precision、召回率Recall、F1-Score
- 实验要求：
  - 训练时只能利用train文件中的数据，优化好后，再进行测试集预测
    - 任务输入：一段情绪文本
    - 任务输出：标签（0或1）
  - 使用实验test.csv来测试训练好的模型。为方便测试，提交的test.csv文件需要按照idx列升序排列。
  - 需要对实验结果进行分析
  - 提交的内容：
    - 代码 + emotion_test.csv + 实验报告
    - 无需提交训练好的模型（包括预训练的模型）

注意事项

- **实验一为必做，其余任选一个**即可
- 最终提交：2份实验报告+2份实验代码+2份预测结果
- 邮件及压缩包命名：**人工智能原理2025课程实验+学号+姓名**
- 发送邮箱：2024110498@mail.hfut.edu.cn

此电脑 › 本地磁盘 (D:) › 人工智能原理2024课程实验+学号+姓名

名称	修改日期	类型	大小
实验二_code	2024/10/31 21:00	文件夹	
实验一_code	2024/10/31 21:00	文件夹	
house_test.csv	2024/10/31 21:01	XLS 工作表	0 KB
minist_test.csv	2024/10/31 21:00	XLS 工作表	0 KB
学号_姓名_实验报告.docx	2024/10/31 21:01	DOCX 文档	0 KB

选择实验一，实验二的提交样例

截止日期12月13日

课程实验评分标准——重要

	评价内容		评价依据	权重
实验验收	评价点E1	实验原理是否理解；代码是否规范，程序能否运行；实验结果是否正确；任务是否全部完成	实验代码 实验结果	0.6
实验报告	评价点E2	报告格式是否规范，语言使用是否规范，行文是否流畅，是否图文并茂	实验报告	0.1
	评价点E3	实验数据记录是否完整，实验结果的分析、对比是否充分	实验报告	0.2
	评价点E4	实验体会是否正确，是否提出了自己独到见解	实验报告	0.1